CIMB NIAGA

Credit Analyst Copilot

Autonomous LOS Reading & Memo Drafting |
One Analyst. Five Loans. Four Minutes.





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Human Bottleneck in Indonesian Banking

Indonesian banks face a capacity bottleneck — not in systems, but in human analysts. Analysts are forced to manually re-enter every loan data field — even when documents are already available.

The irony: all data already exists — analysts just retype it.

30–60 min
per loan (manual entry)

50–60
loans/day target (analyst)

Overtime
required just to hit target

** Operational data from field observation at an Indonesian bank; macro context: OJK, BI, Credable.in, FICO 2024*

Anatomy of the Bottleneck

At 30–60 min/loan, hitting the 50–60 loan/day target means analysts work 25–60+ hours per day — impossible without heavy overtime

Processing Time per Loan

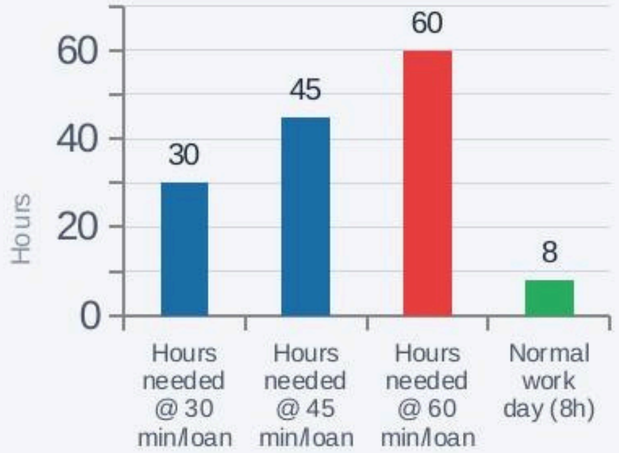
Manual (today) vs. With AI Copilot



Teal bar = With AI Copilot (5–8 min). Blue bars = Manual.

Hours Required to Process 60 Loans/Day

vs. Normal 8-Hour Workday



Red = hours needed for 60 loans at slowest pace (60h). Green = normal 8h workday.

DATA SOURCES & CITATIONS

Field Observation (Primary) 30–60 min/loan, 50–60 loans/day target, mandatory overtime — internal data from interviews with Indonesian bank analysts.

Credable.in / DPI Report (2024) "Manual processes cause slower verification & longer TAT." SMEs receive only 19% of total bank credit in Indonesia.

FICO / RFI Global (2022) "62–70% of Asia-Pacific SMEs dissatisfied with major banks." >67% of Indonesian SMEs seek alternative lending.

OJK / Bank Indonesia (2024–2025) Credit grew 10.39% YoY (2024). Regulations push banks to strengthen risk management & innovation.

Why Important:
The Onboarding & Policy Burden

Rulebook Complexity

15+

credit products, each with its own separate manual policy guideline

Every credit product carries its own evolving guideline. Analysts must master all of them before working independently — with no centralized reference system in place.

→ **Copilot embeds policy rules directly — no memorization needed.**

Knowledge Gap Risk

Slow Analyst Onboarding

3–6

months before a new analyst can work independently at full accuracy

Knowledge transfer relies entirely on senior analysts. When a senior leaves, institutional knowledge walks out the door — there is no structured handover or system to fall back on.

→ **Copilot acts as a persistent policy brain — senior knowledge, always available.**

Operational Continuity Risk

Inconsistency & Audit Risk

19%

of total bank credit reaches SMEs — partly due to inconsistent decisions

Dynamic policies increase the risk of inconsistent credit decisions across analysts. Memo formatting errors or policy misreads can directly trigger OJK audit sanctions.

→ **Copilot flags policy deviations in real-time — every memo, every analyst, same standard.**

Compliance & Audit Risk

Key Takeaway: Without a centralized system, every new analyst introduces new risk. Credit decision consistency depends entirely on individuals — not on process. This is unsustainable at scale. **Credit Analyst Copilot standardizes all three — from day one.**

SOLUTION IMPACT

More Loans. More Revenue.

Increasing potential profit by accelerating your processing rate.

5×

Output per Analyst

50–60 loans/day (with overtime) → 250–300 loans/day (within 8h, no overtime)

Thousands of loan applications arrive daily.

Capacity was the ceiling — not anymore.

94%

FASTER PER LOAN

30–60 min → 5–8 min

Manual data re-entry eliminated entirely. Analysts shift from typists to decision-makers — reviewing AI-structured outputs instead of transcribing documents by hand.

2,750

LOANS/DAY — 10-ANALYST TEAM

550 today → 2,750 with AI (parallel mode: up to 5,500/day)

Same 10 analysts, no new hires. Equivalent to adding ~40 analysts — 50 total capacity at a fraction of the cost. Parallel mode available for surge demand.

Rp 22.9B

TOTAL ANNUAL BENEFIT

Rp 7.2B hiring savings + Rp 16.95B revenue upside − Rp 1.25B AI cost

More decisions per day = more booked loans = direct interest income. At scale across branches, the compounding effect is material and measurable.

No additional headcount. *Same team. Five times the output. Direct revenue impact — from day one.*

The ROI

Payback in Months, Not Years — ~Rp 3.3B Net Annual Benefit (HR Savings), Revenue Upside on Top

Component	Assumptions (illustrative — replace with CIMB data)	Value
Current Team	10 analysts, ~55 loans/day (incl. overtime)	baseline
With AI Copilot	~8 min/loan (down from 30–60 min); conservative 3–4x output, not 5x	~180–220 loans/day
Additional Capacity	Equivalent to ~20–30 analysts (25 used as midpoint)	~25 analysts
HR Savings	25 × Rp180M/year (fully-loaded analyst cost)	~Rp 4.5B/year
AI Cost	Reference Mode B, ~Rp1,800/loan × annual volume	~Rp 1.2B/year
Net Benefit (solid)	HR Savings − AI Cost	**~Rp 3.3B/year**
Payback Period	Light implementation: read-only access, low-CapEx	**~2–4 months**

~Rp 4.5B

Annual HR Savings (Headcount)

~Rp 3.3B

Net Annual Benefit (after AI cost)

2–4 Months

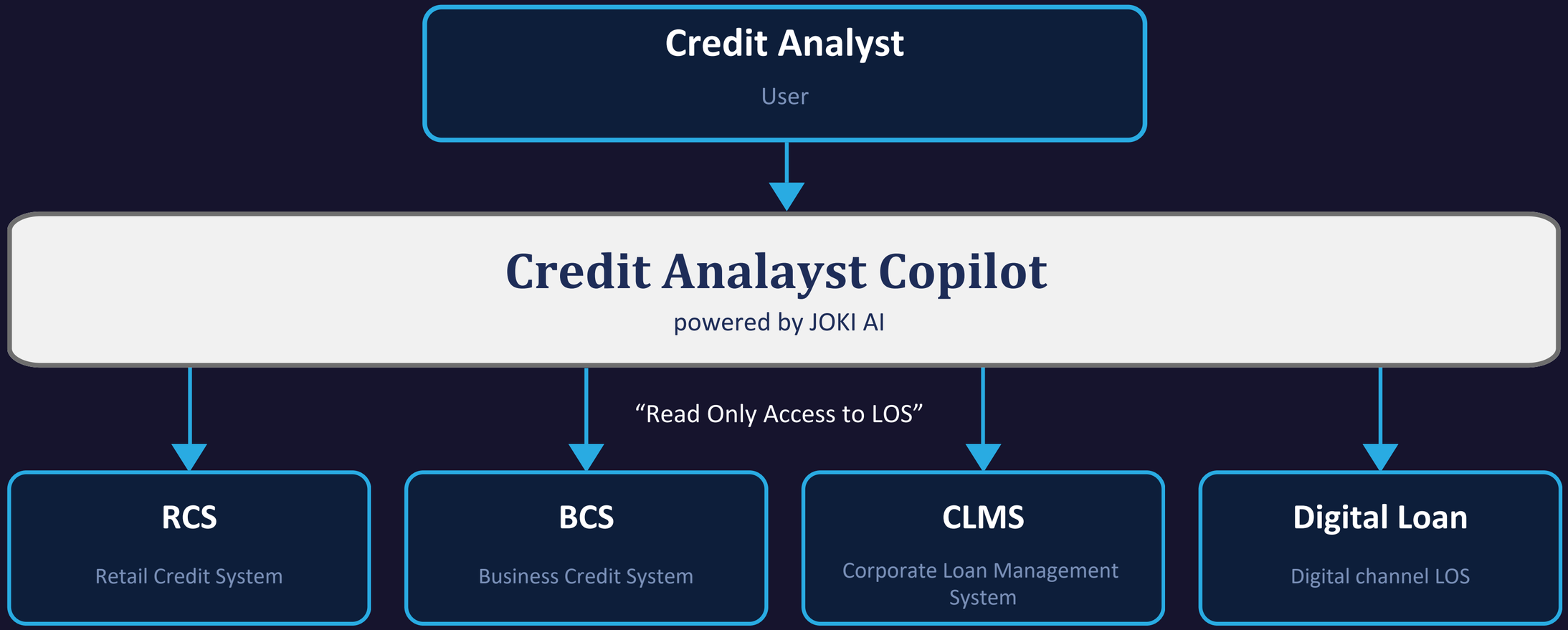
Projected Payback Period

Revenue Upside — Bonus, Not Counted in Payback: faster disbursement could unlock additional interest income on top of this — the size depends on how much extra volume the bank can actually fund and disburse.

** Illustrative assumptions: 10-analyst team, Rp180M/year fully-loaded analyst cost, ~8 min/loan with Copilot (conservative 3–4x output), AI inference cost per Mode B (~Rp1,800/loan). Net Benefit anchored to HR savings only; revenue upside excluded from payback.*

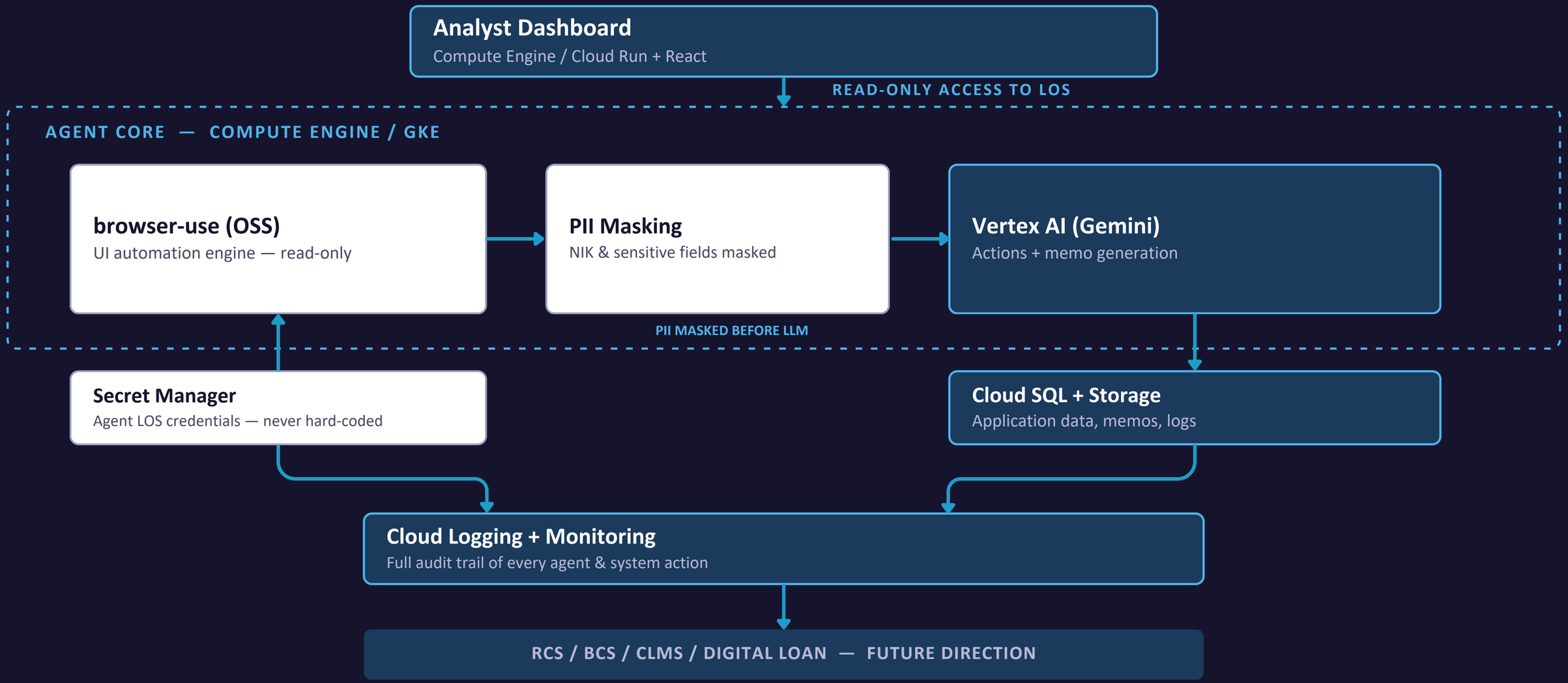
Business Model

Credit Analyst Copilot — agent layer over multiple LOS (Loan Origination Systems)



Deployment Status: Proven on one Dummy LOS today via UI automation. **RCS, BCS, CLMS, and Digital Loan** represent our future integration directions.

Architecture & Security



✓ Read-Only Access to LOS

✓ PII Masked Before LLM

✓ Full Audit Trail

Deployment Cost Analysis

Resolved pricing models, transaction expenses, and scaling projections

DEPLOYMENT MODES & COST MATRIX

MODE	LLM MODEL	EXTRACTION	INPUT/OUTPUT	COST/APP
MODE A	Gemini 3.5 Flash	Browser Agent (Premium)	$1.50/$9.00	**$0.2454**
MODE B	Gemini 3.5 Flash	API Direct (Pro + API)	$1.50/$9.00	**$0.1104**
MODE C	Gemini 3 Flash	Browser Agent (Budget)	$0.50/$3.00	**$0.0818**
MODE D	Gemini 3 Flash	API Direct (Cheapest)	$0.50/$3.00	**$0.0368**

SCALING PROJECTIONS (300 LOANS / DAY)

TIMEFRAME	MODE A	MODE B	MODE C	MODE D
Daily	**$74**	**$33**	**$25**	**$11**
Monthly	**$1,619**	**$728**	**$540**	**$243**
Yearly	**$19,433**	**$8,741**	**$6,478**	**$2,914**

MULTI-ANALYST SCALING (10 ANALYSTS @ 3,000 LOANS/DAY)

Yearly operational costs scale linearly:

Mode A **$194,333** | Mode B **$87,413** | Mode C **$64,778** | Mode D **$29,138**

COMPONENT BREAKDOWN PER LOAN APPLICATION

Browse: $0.1350 (A/B) / $0.0450 (C) Memo: $0.0279 (A/B) / $0.0093 (C/D) Chat: $0.0825 (A/B) / $0.0275 (C/D)

Bottom line: even the priciest mode (Mode A, ~Rp4.4K/loan) is only ~3% of the per-loan analyst cost (~Rp136K). Mode B — our recommendation — runs ~1%, and is the figure carried into the ROI on Slide 5.

WHAT WE BUILT

MVP Deliverables

What's already live at Hackfest, and what's next on the roadmap.

Delivery Confidence

Proven methodology with clear milestones. From hackathon prototype to enterprise deployment in structured phases.

★★★★★ 100%

Done

Built & working at Hackfest

- Backend API + Python agent (core automation engine)
- Batch agent spawning — parallel multi-loan processing
- Dashboard UI — LOS (dummy) + Copilot, fully working
- Memo generation + chat Q&A on the application
- Testing & demo — validated end-to-end

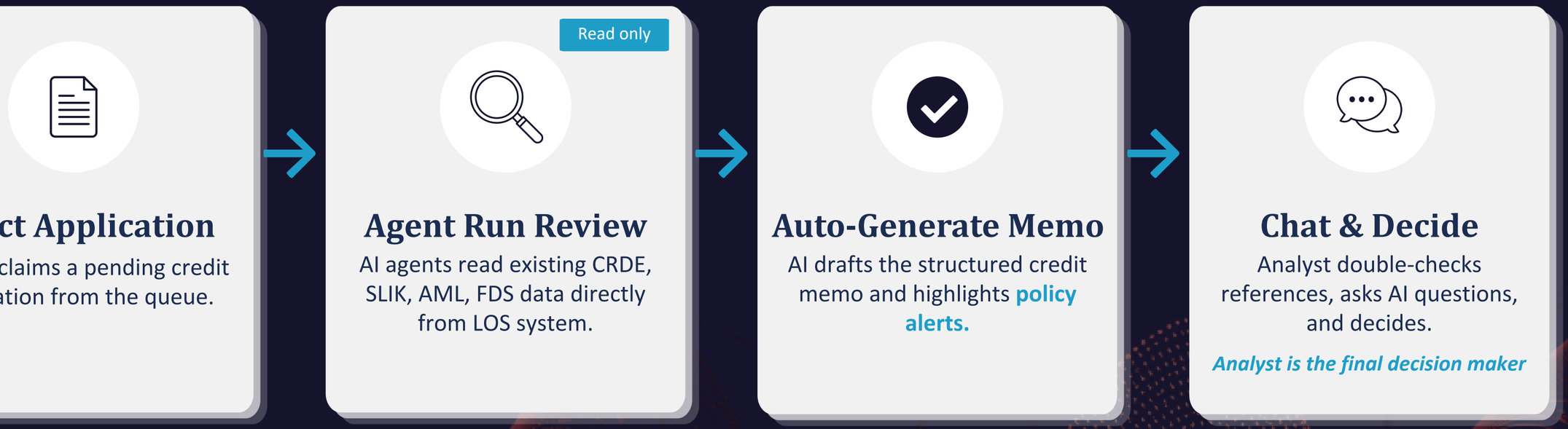
Next

Post-Hackfest roadmap

Phase 1: Real LOS (RCS/BCS/CLMS/Digital Loan), PII masking, SSO
Phase 2: SLIK OJK API, AML integration
Phase 3: Analytics & audit trail
Phase 4: SME loans, multi-branch rollout

AI suggests. Human decides. Approve / Reject / Need More Review is only a suggestion — the final action in the LOS is always taken by the analyst.

Flow Process:
The Automated Underwriting Journey



WATCH IT WORK

Live Demo

WHAT YOU'LL SEE

5 real loan applications — read, summarized, and turned into structured memos in under 5 minutes, live.



LOS — Application View

[drop screenshot / recording here]



Copilot — Memo & Suggestion

[drop screenshot / recording here]

Thank You

Ready to pilot — read-only, low-risk, deployable today.

30–60 min → 5–8 min

94% faster, per loan

5× output

per analyst — no new headcount

2–4 months

projected payback period